



## Razer's new Xbox 360 controller

Razer launched a new controller for the Xbox 360 called the "Razer Sabertooth". It has a built-in OLED Display along with six extra buttons



## 18 Billion messages

WhatsApp, the popular instant messaging app processed a record 18 billion messages on Dec 31, 2012

silent film era and we learned to appreciate the inherent blur that the frame-rate introduced. Directors also used this to full effect and the blurring, coupled with the softer images of the non-HD era produced movies that looked dreamy and surreal. This worked exceptionally well for movies such as *Casablanca* or *Citizen Kane*, but fell short of completely immersing audiences in proper action movies such as *The Towering Inferno* or *The Burning Train*. The inherent blurring ruined fast-paced scenes, but since the audience were used to it, it didn't matter that much.

The argument against HFR is the same that was used against HD. People complained that HD brought with it more detail than was required and would be the end of the "close-up shot" and there would no longer be any romantic (and arduously long) kissing scenes. This isn't the case though and any good director knows that it is softer focus, not a lower resolution that brings that effect.

## To blur or not to blur

It isn't possible to talk about HFR and immersive movies and not mention Douglas Trumbull. He was the man who produced special effects for such iconic movies as *Blade Runner*, *Star Trek: The Motion Picture* and even *2001: A Space Odyssey*. Trumbull was a pioneer, he wanted to revolutionise film-making in general and was a great proponent of the HFR format. His work on special effects is still unmatched and it is only current generation CG that has bettered some of his work in this field.

What Trumbull wanted was a movie that was as immersive as possible, a first-person experience, a movie that was your own where you were part of the movie rather than a disconnected member of the audience. To this end, 24 fps wasn't enough, a higher frame-rate was required and 48 or 60 fps was the key. At this frame-rate, blur disappears, a well-shot video at 60 fps, projected on a screen of sufficient quality, is as close to being indistinguishable from reality as it is possible to get. Incidentally, this is why 60 fps is the holy grail for a gamer. 25 fps is playable, sufficient, but not ideal and certainly not recommended.

When it comes to blurring or a loss of detail, be it in an image or even audio, our brain has the impressive ability of filling in the details, a sort of interpolation of data. This is why a blurry CG image can look way better than a crystal clear CG image if that CG image isn't as detailed as it should be. It is this ability to interpolate and interpret incomplete data that directors inadvertently (the good ones do it deliberately) take advantage of when they're shooting their scenes. HFR



Blur is not always a bad thing

gets rid of this factor and presents the viewer with exactly what has been shot, leaving no room for interpretation. It's like looking at the movie through a window and everything is actually happening in front of your eyes. Unless what you're seeing is "real" or as close to reality as it's possible to get, what you see through that "window" will seem fake, just a bunch of actors on a stage, playing with props.

The challenge faced by the director of *The Hobbit* was exactly this. His "window" wasn't real enough to immerse the audiences. People complained that props looked like props, CG sequences looked like game trailers and some people absolutely loved the fast-paced action sequences. Peter Jackson failed to live up to the expectations of HFR, but at least he tried and while the technology may not have been well implemented, the movie seems to have done an excellent job of show-casing the good and bad of that technology.

To upgrade to HFR requires projectors that are capable of projecting the movies at such speeds, but the fact is that most good projectors are perfectly capable of handling the task and many of them routinely project at over 72 fps anyway (144 for 3D).

## 3D and HFR

What happened with *The Hobbit* is what has been happening with 3D. Directors haven't yet fully wrapped their heads around the technology so while *Avatar* looked fantastic in 3D, *Clash of the Titans* looked awful. What is required is a complete reimagining of every scene in 3D and HFR, which is actually quite hard for someone who has had to do the exact opposite their whole life. Shooting in 3D or HFR is not just about shooting

the scene with fancy gear for the sake of shooting it. The entire visual experience is defined by the techniques that are used and adequate care must be taken. HFR and 3D tech have been around for decades, they just haven't been implemented well enough for mainstream use.

There is a lot of talk that "3D is dead" or that it was a huge mistake. There is similar talk about HFR as well and this just after one movie. This is a technology that hasn't been given a chance to mature yet. The potential is there and these technologies are the future, all that is lacking is the right vision and a better understanding of the technology. When it comes to movies, these technologies are not just mere jargon that can be thrown around to attract audiences. They can affect the entire movie experience and must be used with care and precision to truly take advantage with them.

Just like an artist has brushes and paints at his disposal, HFR and 3D are tools that a director can employ. These tools need to be used judiciously and with care. What is the point of an illusion if you can clearly see what the magician has employed to hide it in the first place?